

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (IT)

March-April 2018

IT202: Data Communication

Date: 06.04.2018, Friday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Write the answer of following Questions.

- (a) Compare following Series tuned circuits, Parallel tuned circuits. [06]
- (b) What is mutual inductance? [02]
- (c) What are the applications of transformer? [02]
- (d) What do you mean by high frequency transformer? [01]

Q - 2 Write the answer of following Questions.(Attempt Any Two)

[12]

- (a) Explain Capacitive Tap.
- (b) Compare different types of transformers.
- (c) Draw and explain Aperiodic Signal.

Q - 3 Write the answer of following Questions. (Attempt any TWO)

[12]

- (a) Explain properties of Fourier transform.
- (b) Describe signal power and spectral power density.
- (c) Compare signal power and signal energy.

SECTION – II

Q - 4 Write the answer of following Questions.

- (a) Define Modulation. [01]
- (b) Explain frequency synthesizer. [06]
- (c) Compare Amplitude modulation vs. Frequency modulation. [04]
- (d) What do you mean by Non sinusoidal Modulation?

Q - 5 Write the answer of following Questions. (Attempt any TWO)

[12]

- (a) Explain Cascode Amplifier.
- (b) Explain Crystal Oscillator.
- (c) Explain Double-Sideband Suppressed Carrier.

Q - 6 Write the answer of following Questions. (Attempt any TWO)

[12]

- (a) Explain Electronically tuned receiver.
- (b) Explain noise in AM system.
- (c) Explain Demodulator Circuit.
